Nubra · Product Management Intern Assignment

Habit Ledger

A post-trade review that only names habits it can defend

Problem 03 · Post-trade review

Working application

https://habit-ledger-xi.vercel.app

Source code, tests and documents

https://github.com/vinayakarya02/habit-ledger

Contents of this submission: 1. Product one-pager · 2. Prompt log · 3. Evaluation report · 4. Working application

Submitted 23 October 2026. All trader data in the app is synthetic; the prototype is not connected to Nubra or any broker and gives no investment advice.

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1. Product one-pager

Habit Ledger: a post-trade review that only names habits it can defend

Problem 03 · Post-trade review · App: https://habit-ledger-xi.vercel.app · Code: https://github.com/vinayakarya02/habit-ledger

My take on the problem

"Turn a month of order history into the two or three habits that are costing them money" contains three tests that most post-trade tools fail:

- **"Costing"** needs a counterfactual. About 91% of individual F&O traders lost money in FY25 (SEBI, July 2025), so nearly every bucket of their trades loses. Subtracting losing trades always "finds" a cost. The honest question is how much *worse* these trades did than the trader's usual trade.
- **"Habits"** means repetition. A month is 30–200 trades, and one blow-up can impersonate a pattern. A credible review must be willing to say "not enough evidence".
- **"Them"** means a personal rule the trader can act on, not generic tips.

A later SEBI study found loss rates *rise* with experience. Experience alone doesn't correct behaviour, so the review must close a loop.

Why Nubra should solve it

Nubra holds both ends of that loop: the tradebook and P&L reports in its public API, and Flexi and strategy orders with stop-losses and whole-strategy exits. Nothing public connects a trader's history to *which* lever they need. Competing reviews (e.g. Upstox Trading Insights' expiry and call/put buckets) describe where money went. They don't test whether a behaviour is real, price it, or turn it into a rule. Nubra's multi-leg users also show a habit generic tools miss: closing the hedge and keeping the naked short.

Solution space

I compared descriptive dashboards, AI-written journals, rule-break ledgers (cold start), market-price replays (need intraday data, invite hindsight), and **evidence-gated habit detection with a guardrail loop**. Only the last is personal, falsifiable and actionable from a tradebook alone.

What I built

Five behaviours are checked, using only what the trader knew when deciding:

- re-entering within 15 minutes of a loss;
- adding to a losing option buy;
- removing a spread's hedge;
- sizing up after two wins;
- trading past the 5th trade of a day.

Cost is the excess loss versus the trader's own usual trade, capped so it never counts forgone profit. A behaviour becomes a **habit** only if:

- it occurs at least 4 times on at least 3 days;
- it is material;
- it still matters without its single worst trade;
- the period has 20 or more trades.

A disciplined trader gets **zero** warnings.

Nine connected modules make that reviewable:

- an **Overview** with one next action;
- **Habits**, with the reason for each classification;

- an **Evidence inspector** showing the trades, baseline, arithmetic, leave-one-out, and a re-gate when you exclude trades;
- a **Trade explorer**, where missing values show as "—", never ₹0;
- **Guardrails**, each labelled as a documented Nubra order control or a proposed capability, with compliance measured from fills;
- **Compare periods**, using rates per chance and an exact test, so trading less never counts as improving;
- **CSV import** with column mapping, validation and duplicate checks; data stays in the browser;
- a **Scenario lab**;
- a **Methodology** page.

140+ automated tests check planted behaviours against ground truth, recompute costs independently, test threshold boundaries and false-positive traps, and render every view across ~90 states. Bugs found by adversarial review became tests.

In scope

- Closed option trades (single- and multi-leg) from CSV or synthetic sample traders, reviewed month by month.
- No AI at runtime, by design: deterministic maths keeps every number auditable.

Out of scope

- Live Nubra connection, credentials or order placement.
- Nubra's own tradebook format, whose fields aren't public.
- Market-price counterfactuals.
- Equity, futures and tax.
- Per-user thresholds, which would invite fishing for patterns.
- Any trade recommendation.

What would prove me wrong

- Traders don't adopt guardrails.
- Guarded habits' rates fall no more than unguarded ones'.
- Traders dispute the evidence.

A launch would track all three, plus the share of zero-habit reviews (near 0% means the gates are loose) and order counts after review, which must not rise.

2. Prompt log

These are my prompts in order, written in shortened form. Prompts 1–3 and 6–10 come from two longer messages, broken into their steps. The full text is in 06_prompt_log.md.

Prompt 1

Research Nubra's public product before we build anything. Go through its website, product pages, Play Store listing, API documentation and release notes. Tell me what it already offers, which kinds of traders it serves, and what problems its current features seem designed to solve. Also tell me where an incremental copy of an existing feature would add little value, and where real unmet needs might remain. Keep verified facts separate from inference, cite sources, and don't invent internal features, customer interviews or metrics.

Prompt 2

For each of the four problem statements, explain the problems traders face today and where Nubra could improve. Then suggest product ideas for them, saying who each is for, when the problem happens and how it differs from what Nubra already has, and score them.

Prompt 3

Could Nubra already have it? Is it just a feature dressed up as a product? Is AI really needed, or would plain logic work better? Then give me your recommendation.

Prompt 4

We are creating a product called Habit Ledger. It is a post-trade review for active options traders: it takes a month of their orders and finds the two or three repeated behaviours that are costing them money. It should only call something a habit when the evidence holds up, show the exact trades behind every finding, and estimate the cost honestly against the trader's own usual trade. Each habit should map to a personal guardrail, and the next month should show whether the behaviour actually changed. Keep all detection and calculations deterministic. Create sample traders, including a disciplined one who should get no warnings. Then build it and test it.

Prompt 5

Deploy the app to Vercel as a public site and check it opens without logging in.

Prompt 6

Now turn Habit Ledger into a complete product without changing the core idea. Check the current app first so nothing that works gets lost. Then add an overview page that answers what happened this month, which habits have enough evidence, what changed since last month, and what the trader should do next. Also add a habits section that shows how often each behaviour happened out of its chances, the trades behind it, and why it was or wasn't classed as a habit.

Prompt 7

Add a trade list with search, filters and a detail view for every trade, showing missing values as missing, never as zero. Then add an evidence page for each habit with the exact trades, the comparison baseline, how the cost was calculated, and what happens without the worst trade.

Prompt 8

Add a guardrails section that turns a habit into a personal rule, like a cool-off after a loss or a daily trade limit, and mark which rules Nubra already supports. Then add a month comparison that uses rates instead of raw counts, so trading less doesn't count as improving.

Prompt 9

Add CSV import with column mapping, a preview and clear errors, and never turn bad values into zero. Add a scenario lab clearly labelled as a simulation, and a methodology page that explains every rule.

Prompt 10

Add tests for the edge cases and fix anything that breaks. Then redeploy to the same link and check that the live app matches the code.

3. Evaluation report

Habit Ledger: Evaluations

How I decided it was good enough to ship (as a prototype):

1. It finds behaviours that are really there.
2. It stays silent when they aren't.
3. Every rupee figure can be recomputed by hand.
4. Bad input is rejected, never silently turned into ₹0.
5. Every control in the UI does what its label says.
6. Independent reviewers who are trying to break it can't find a confirmed problem that remains unaddressed.

Actual result (23 Oct 2026, Node v24.16.0):

- `npm test` passes **151 / 151** (0 fail, 0 todo, 0 skipped).
- `npm run smoke` renders 9 views × 90 states = **810 renders, 0 problems**.
- `npm run build` succeeds.
- The raw test output is saved in `app/tests/last-run.txt`.

All figures in this document come from actual runs.

1. Ship criteria

#	Criterion	Threshold	Result
S1	Detection against planted ground truth (5 behaviours × 6 sample trader-months, plus lab scenarios)	Detected set = planted set	Met (E01–E03, lab tests)
S2	The disciplined trader gets no confirmed habit	0 in both months	Met (E04, U-tests)
S3	Each near miss fails for the intended reason	3 / 3	Met (E05)
S4	Costs equal an independent recomputation from raw fills	abs. error < ₹0.01	Met (E07, E08, engine rules)
S5	Invalid values are rejected with a reason, never coerced to 0	All malformed cases	Met (E10–E12, import tests, R13)
S6	No copy claims causation, forgone profit or advice	0 matches across all rendered states	Met (E17, U07)
S7	Verdicts follow the comparison rules, and trading less never counts as improving	All boundary cases	Met (E14, E15, compare tests)
S8	Every rendered control maps to a real action; no duplicate IDs; no leaked `NaN` or `undefined`	0 problems in 810 renders	Met (U01, smoke)
S9	Every control has an accessible name; one h1 per view; charts have text alternatives	All sampled states	Met (U03, U04)
S10	Confirmed adversarial-review findings addressed	57 / 57	56 fixed, 1 partly fixed (see §4.3)

2. Automated suites

File	Tests	What it covers	How expected values were derived
`evals.test.mjs`	19	The original v1 evals: planted vs detected, false-positive traps, cost arithmetic, gates, CSV failures, open positions, round trip, verdicts, de-duplication, honest copy, thin month, data realism	Planted manifest from the generator; a hand-worked spread; recomputation from raw fills

File	Tests	What it covers	How expected values were derived
`engine.rules.test.mjs`	34	Every detector at its exact threshold boundary, every gate boundary, baseline cap and thin baseline, zero and flat P&L, missing charges, overlap labels, period split, guardrail measurement with custom parameters, re-gate, the 3-habit limit	Small hand-built fixtures, with the expected figures worked out in comments
`import.test.mjs`	27	Reader (quotes, CRLF, BOM, `;` delimiter, line numbers), header aliases, mapping problems, timestamps (zones, no zone → IST, invalid dates, seconds ≥ 60), strict amounts (Indian and western grouping; decimal comma rejected), every row-level error, duplicates, the row limit, round trip	Literal inputs and outputs
`compare.test.mjs`	16	Fisher exact against hand counts (17/70, 1/6, 2/19, 1/26) and an exact reference; Wilson; every verdict boundary (minimum chances, ±50% / 25% relative, p on both sides of 0.2 and 0.05); "trading less is not improving"	Exact fractions
`lab.test.mjs`	10	Each preset's classification, determinism, gap outside the window, clamping, planted = detected	Generator manifest
`store.test.mjs`	23	Reducer purity and state resets (trader or period change resets filters, selection, exclusions and drafts); ignored bad input; guardrail flow; import flow; persistence sanitising; next-action priority; trade filters and paging; inspector sensitivity	Hand-checked counts from sample data
`ui.test.mjs`	9	Render matrix (no unknown actions, duplicate IDs or leaks); every effect implemented; accessible names; one h1; chart text alternatives; overview headline = engine headline; no stale data after switching trader; disclaimer present and banned phrases absent; overlays only when asked for; missing values shown as "—"	Rendered HTML
`review-fixes.test.mjs`	13	One regression test per engine or store fix from the integrated review (R01–R13, §4.2)	The reviewers' reproductions

The render matrix (`tests/helpers/render-matrix.mjs`) covers 90 states:

- 3 sample traders × 2 months × each habit in the inspector;
- trade filters, selections and dialogs;
- every import-wizard step, including a messy broker file and a thin file;
- every lab preset plus edge settings;
- three edge datasets: a habit cut only by the 3-habit limit, and two real overlaps.

3. Changes to existing tests (all deliberate)

Test	Change	Why
E15	Expected labels renamed: "Worse" → "Worsening", "Too little activity to judge" → "Inconclusive"	v2 verdict vocabulary. The same cases now go through a stricter rule (minimum 8 chances, exact test).
E17	Scans every file under `src/` instead of two files	The v1 `app.js` no longer exists. The scan is wider, not narrower.
E19	Days of Meera's hedge removals read from the ground-truth manifest	The sample calendar now skips NSE holidays, so the hard-coded dates moved.

Test	Change	Why
engine.rules "previous trade flat"	Re-entry chances 1 → 0	**Semantic fix** (review finding #1). A re-entry chance is now a *trade opened after a losing trade*. Before, chances counted losing trades while occurrences counted trades, so the rate could exceed 100%.
compare "costs floored at 0"	Chances label → "trades opened after a losing trade"	Same fix
lab and store "days" clamp	22 → 21	September 2026 has 21 NSE trading days
store selectDataset	Inspector opens on the new trader's top confirmed habit (H3 for Meera) instead of keeping H2	Review finding: switching trader showed an empty "not seen" page
store guardrail, persistence and import tests	Rule objects gained `settingsPeriod`, `savedAt` and `changedAt`; rejected rows gained `file`	New fields for the honest guardrail timeline and multi-file imports
store next-action	October with rules set → "review-change" (compare) instead of "wait"	Review finding: the month where the loop pays off was a dead end
store import notes	Timezone note counts 4 imported rows, not 6 rows read	Review finding: it counted rejected rows that never entered the data

No test was deleted, and no assertion was loosened to make a failure pass.

4. Adversarial review (real, multi-agent)

4.1 Module build workflow

Ten pipelines ran (nine modules and one test author), each through build, independent review and fix: 30 agents in total. The test author found **6 real bugs** and recorded them as `todo` tests instead of patching them:

1. A decimal comma "112,50" was read as ₹11,250.
2. A timestamp with seconds ≥ 60 became `NaN`.
3. The import notice double-counted excluded rows.
4. `initialState(null)` crashed.
5. The p ≥ 0.5 sentence was ungrammatical.
6. A 4th habit cut only by the 3-habit limit was called an "overlap".

All six were fixed, and the tests are now hard tests.

4.2 Integrated review

Five reviewers each took one lens:

- numbers and cross-module consistency;
- interaction and state;
- honesty and copy;
- accessibility and responsive design (in a browser);
- the product as a whole.

Two independent skeptics then tried to refute each finding.

	Count
Findings reported	66
Confirmed by both skeptics	**57** (1 critical, 24 major, 32 minor)
Split (one skeptic each way)	4
Rejected by both	5

The most important confirmed findings, and what changed:

- **Critical: the re-entry rate could exceed 100%.** Two trades opened after one loss both counted against a single "chance". This made compliance negative, gave a `NaN` interval and produced a false "strong Improving". A chance is now a trade opened after a losing trade, and the statistics functions reject impossible inputs (R01, R02).
- **Honesty: guardrail timeline.** Months already in the file were labelled "after the rule was set", implying the rule caused a change. Each rule now records when its settings were saved. A month is "after" only if its trades came later; otherwise it is labelled a what-if. Compare no longer says a rule was "broken" before it existed (R06).
- **Honesty: suggested stop levels.** The stop-loss snippet showed specific stop distances (0.70 / 0.68 × entry), and the adding-to-a-loser rule was labelled "documented" although no documented control blocks adding. The rule is now only the measured behaviour, labelled proposed. The snippet carries field names with placeholder values (R05).
- **Honesty:**
  - "Mostly the same trades" was used for 1 of 19 shared trades (R03);
  - limit cuts were called overlaps (R04);
  - "gross losses" actually included charges;
  - the headline said "vs your usual trade" for figures measured against ₹0 or a spread's maximum loss;
  - sample trades fell on NSE holidays (R11);
  - Kabir was described as using stops;
  - counts were not pluralised.
- **Product:**
  - "Import next month" was impossible: each import became a new trader. A file can now be added to an existing imported trader (R07).
  - The inspector had no way to set a guardrail.
  - Suggestive changes looked as certain as clear ones. They now read "Possibly improving", and behaviours that were never a habit are shown as context (R12).
  - The inspector opened on an empty habit (R10).
  - A clean trader was told to wait for a month already loaded.
- **Interaction and accessibility:**
  - slider drag broke after one step;
  - dialogs didn't trap focus;
  - Escape closed the dialog instead of the popover inside it;
  - re-renders reset scroll positions;
  - the skip link broke routing;
  - the live region was never written to;
  - the focus ring was hidden on active segments;
  - text contrast was slightly below AA, and input borders were 1.55:1;
  - popovers overflowed the viewport;
  - the chart's values were reachable by hover only;
  - the phone header took 43% of the screen.

Split findings:

- Implemented:
  - the loop-payoff next action (now "review-change");
  - the lab → importer handoff ("Open this month in the importer");
  - hiding another trader's data-quality panel while a file is checked.
- Not implemented: larger tap targets for the 16px "?" popover triggers. One skeptic rated this not-an-issue, the other minor.

Rejected by both skeptics: five findings, including that lot-cap compliance measures only the first fill (it measures entry size by design), and that the Overview leads with generic KPIs.

4.3 Fix verification

Engine, store, main and data fixes were made directly. Each has a regression test (R01–R13).

View fixes went to nine agents, one per module. An independent verifier per module checked every item against rendered text:

- all **9 / 9** were verified as resolved;
- four verifiers found small regressions introduced by the fixes (e.g. two live-region sources on one page, a missing space between sentences, copy that over-claimed), and a second pass fixed them.

Partly addressed: review finding #16. Fixing the unprompted re-render (the notice timer) is done. But any re-render still closes open `<details>` disclosures, because the app re-renders the whole tree.

5. Manual browser QA (23 Oct 2026, local preview, Chromium)

Check	Observed
Guardrail from the Overview "Next step" with a real click	Dialog opens with focus on the first setting; the page behind is `inert` with scroll locked; 8 Tab presses stay inside the dialog; arrow keys change the setting with a live preview; one click on Save closes the dialog, focus returns to the next-step button, and the live region announces "Saved … Nothing was sent to any broker."
Guardrails and Compare wording after saving	"Every period here is a what-if…"; Compare: "9 / 30 → 2 / 24 · Improving · … about 1 in 20 times by chance"; excess loss shown as context only
Trades: typing "BANKNIFTY" in search	Focus stays in the box through each re-render; "Showing 18 of 52 trades · net −₹7,120" announced
Trade drawer	Opens with focus on its heading; Escape closes it and returns focus to that row's Details button
Import of a messy broker CSV via the real file input	Header aliases mapped. Rejected with reasons: decimal comma, duplicate ID, FUT row, side "HOLD", negative quantity, blank price, non-numeric charge. The blank charge was imported and flagged. "1 position will be left out" is named before import. The new trader is thin, so the next step is "Add more history".
Lab slider: real mouse drag from 18 → 5	Value follows the drag to 5; the results recompute; focus stays on the slider
375px in dark mode, all 9 modules	0 horizontal overflow; the current tab is visible in the mobile tab bar; the drawer is a bottom sheet and the dialog fits
Console	No messages at all across the journey

Found in earlier browser QA and fixed: a 1-trade upload was reading as "no habits" (now "Too few trades to judge"), and the copy said "Some appeared" when nothing had.

6. What these evaluations do *not* establish

- **Real-world accuracy.** The data is synthetic and the behaviours were planted. Perfect detection shows the rules are implemented as defined. It does not show that real traders' habits look like this, or that the thresholds are right. Next step: consenting users' real tradebooks, with their own "that's me / that's not me" labels.
- **Causation.** Excess loss is an association; the UI says so wherever a cost appears.
- **Behaviour change.** October was also generated. The loop shows the *measurement* works, not that guardrails change behaviour, which would need a controlled comparison.
- **Statistical calibration.** Gates are a pragmatic small-sample filter. Testing five behaviours raises the chance of a false pattern. A null simulation by a reviewer showed a directional verdict about 10% of the time per behaviour per direction with the $p < 0.2$ bar, which is why such verdicts read "Possibly…".
- **Nubra integration.** Nubra's real tradebook schema isn't public. "Documented" labels refer to Nubra's public API docs, not verified app behaviour.

- **Accessibility.** Automated checks plus keyboard tests are not a full audit. Screen-reader testing on real assistive technology wasn't done.

7. AI in the product

There is none at runtime. Detection, costing, gates, statistics, verdicts and compliance are deterministic, so tests can assert exact values. Two places an LLM could earn its keep later, each with structured output, validation and a fallback:

1. **Mapping arbitrary broker CSVs.** Validate by zero row errors plus reconciliation with the broker's P&L. Fall back to the manual mapper, which exists today.
2. **Rephrasing habit cards.** Every number in the text must appear in the structured habit object. Fall back to template copy.

4. Working application

Live app: https://habit-ledger-xi.vercel.app

Repository: https://github.com/vinayakarya02/habit-ledger

The app opens on a synthetic sample trader with no sign-in. It runs entirely in the browser: no account, credentials or market data are needed, and imported CSV files are never uploaded.

Deployment check (23 Oct 2026)

Check	Result
Public URL without login	https://habit-ledger-xi.vercel.app returns HTTP 200
Deployed build	`/version.json` reports commit `2747181afe5f`, the app source commit on GitHub `main`
Hosting	Vercel static deployment (project `habit-ledger`)
Tests on that commit	`npm test` 151 / 151 pass; `npm run smoke` 810 renders, 0 problems; build succeeds

Try it in two minutes

1. Overview (sample trader Aarav, September 2026): 52 closed trades, net −₹1,01,510; two habits pass the evidence bar.
2. Open "Inspect evidence" on "Re-entering right after a loss": the 9 qualifying trades, the baseline, the arithmetic and the leave-one-out test.
3. Click "Set guardrail", choose a cool-off time and save. Nothing is sent to any broker.
4. Open "Compare periods": the re-entry rate falls from 9 of 30 in September to 2 of 24 in October (Improving), with sample sizes and 90% ranges.
5. Switch the trader to Kabir: no confirmed habits, only near misses with the reason each fell short.
6. Open "Import CSV" or "Scenario lab" to see invalid rows rejected with reasons, never read as ₹0.

Supporting documents in the repository

Document	What it contains
01_research_and_concept_selection.md	Research on Nubra and competitors, 14 concepts across the four problems, scoring and the choice of Habit Ledger
02_refined_mvp.md	Product specification as shipped: behaviours, gates, comparison, guardrails, modules
03_implementation_prompt.md	Build specification and the multi-agent workflow scripts used during the build
README.md	Architecture, how to run the app and tests, limitations